Yuxin Yang

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RESEARCH PROFILE

Second-year Ph.D. student in the Department of Computer Science and Engineering at the University of California, Santa Cruz, advised by Prof. Yuanchao Xu. Research interests include secure memory systems, CXL-based memory disaggregation, trusted execution environments, learned indexes, and distributed file systems.

EDUCATION

University of California, Santa Cruz

Ph.D. in Computer Science Aug. 2025 - Present

Advisor: Prof. Yuanchao Xu | Research: computer architecture and secure CXL memory systems

Huazhong University of Science and Technology

M.S. in Computer Architecture Sept. 2022 - June 2025

Advisors: Prof. Fang Wang and Prof. Dan Feng | Average score: 87.3/100

Huazhong University of Science and Technology

B.E. in Computer Science Sept. 2018 - June 2022

Average score: 88.9/100

PUBLICATIONS

[1] ALT-index+: An Efficient Hybrid Learned Index for Concurrent Memory Systems

Mengya Lei, **Yuxin Yang**, Fang Wang, Zhiwei Ye, Dan Feng.

ACM Transactions on Storage (TOS), 2026. **Accepted.**

[2] Decoupled-DSM: A Scalable Secure Trusted Execution Environment for Multi-Host CXL Disaggregated Shared Memory

Yuxin Yang, Yuanchao Xu.

59th IEEE/ACM International Symposium on Microarchitecture (MICRO), 2026. **Accepted.**

[3] ALT-index: A Hybrid Learned Index for Concurrent Memory Database Systems

Yuxin Yang, Fang Wang, Mengya Lei, Peng Zhang, Dan Feng.

41st IEEE International Conference on Data Engineering (ICDE), 2025.

[4] Low-Latency and Scalable Full-path Indexing Metadata Service for Distributed File Systems

Chao Dong, Fang Wang, **Yuxin Yang**, Mengya Lei, Jianshun Zhang, Dan Feng.

41st IEEE International Conference on Computer Design (ICCD), 2023. **Best Paper Nomination.**

RESEARCH EXPERIENCE

Secure CXL memory systems

University of California, Santa Cruz

- Decoupled-DSM provides a trusted execution environment for multi-host CXL disaggregated shared memory; accepted to MICRO 2026.
- AIORE studies adaptive incremental offloaded (re-)encryption for secure CXL memory; presented at NVMW 2026.

A hybrid index structure for memory systems

Huazhong University of Science and Technology Oct. 2022 - Apr. 2025

- Introduced ALT-index, a hybrid learned index for concurrent memory systems (ICDE 2025).
- Improved performance by up to 1.9x-2.3x compared with existing indexes.

A resizing scheme for file systems

Huazhong University of Science and Technology Oct. 2023 - Dec. 2024

- Designed a head-shrinking scheme for a flash-friendly file system, reducing average resizing time from 60 to 15 minutes.

Low-latency metadata service for distributed file systems

Huazhong University of Science and Technology Mar. 2022 - Sept. 2022

- Proposed Duplex, an indexed metadata service, with performance improvements of up to 7.6x / 2.3x over existing solutions (ICCD 2023, Best Paper Nomination).

TEACHING EXPERIENCE

Teaching Assistant, CSE 121: Embedded System Design

University of California, Santa Cruz Mar. 2026 - June 2026

ACADEMIC SERVICE

Reviewer, IEEE Transactions on Computers

SELECTED TALKS

AIORE: Efficient Security Support for CXL Memory through Adaptive Incremental Offloaded (Re-)Encryption
NVMW 2026 Mar. 9, 2026

ALT-index: A Hybrid Learned Index for Concurrent Memory Database Systems
ICDE 2025 May 21, 2025

HONORS & AWARDS

MICRO Student Travel Grant 2026
Best Paper Nomination, ICCD 2023

TECHNICAL SKILLS

Programming: C, C++, Python, Java